Softwarica
COLLEGE OF IT & E-COMMERCE

in collaboration with

Coventry University

Module Name

ST6005CEM Security

CYBER SECURITY | RISK | PROTECTION



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SECURE TODAY | SAFER TOMORROW

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Certificate evidence

Figure 1 shows the certificate presented by the server as seen in the browser Certificate Viewer.

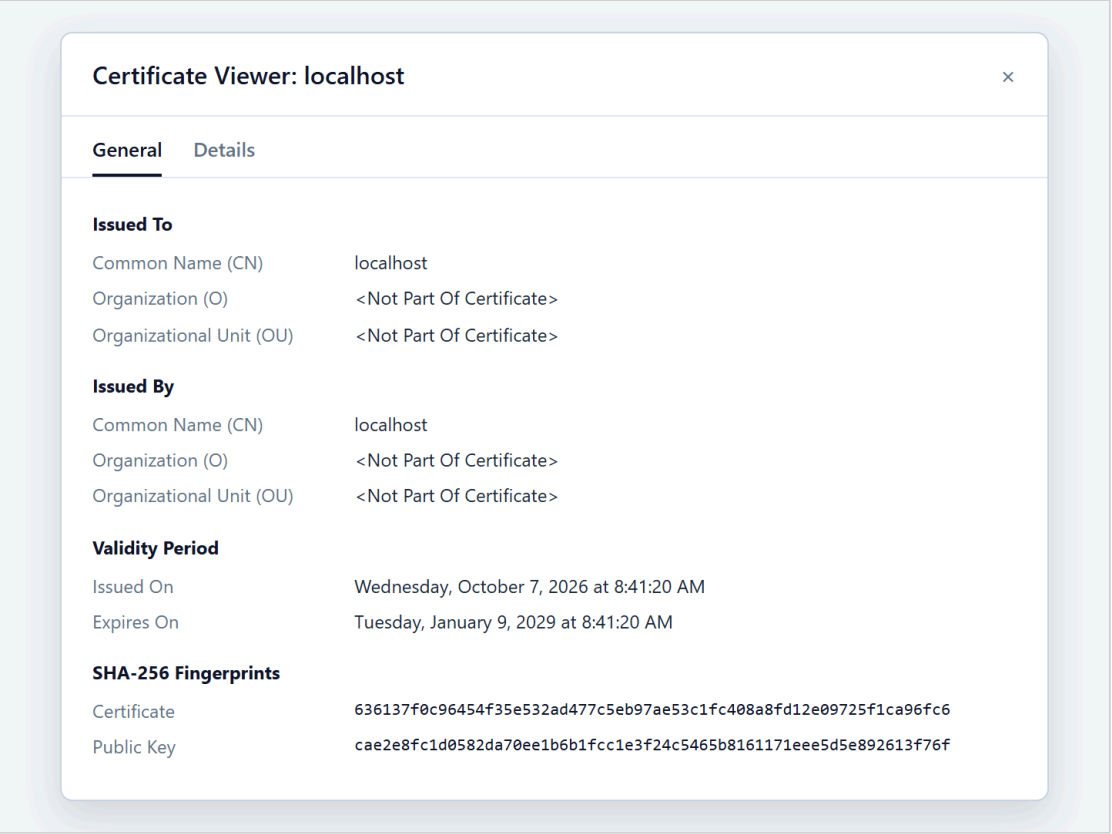


Figure 1: Certificate

Field	Value
Common Name (CN)	localhost
Issued by	localhost (self-signed)
Valid from	Wednesday, 7 October 2026, 8:41:20 AM
Valid until	Tuesday, 9 January 2029, 8:41:20 AM
SHA-256 fingerprint (certificate)	`636137f0c96454f35e532ad477c5eb97ae53c1fc408a8fd12e09725f1ca96fc6`
SHA-256 fingerprint (public key)	`cae2e8fc1d0582da70ee1b6b1fcc1e3f24c5465b8161171eee5d5e892613f76f`

Frontend Implementation

The frontend consists of a signup page and a login page. Both are loaded from https://localhost:3443 and submit to relative HTTPS endpoints, so the browser cannot downgrade to HTTP.

Signup page

The page collects username, email and password. The password field is masked and states the 12-character minimum. Invalid emails are rejected by the server.

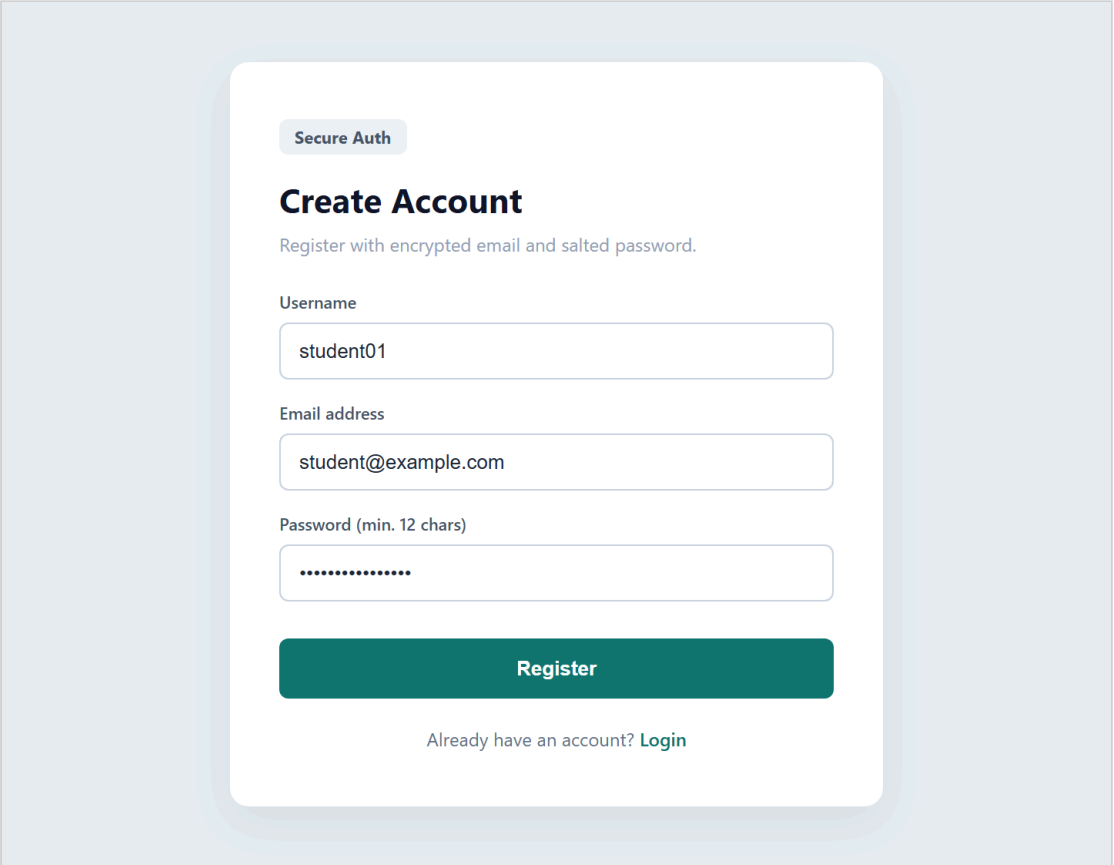


Figure 2:signup page

Filter: `tcp.port == 3443`

No.	Time	Source	Destination	Protocol	Length	Info
114	23.580477000	127.0.0.1	127.0.0.1	TCP	44	3443 → 55860 [ACK] Seq=1345 Ack=3159 Win=62208 Len=0
119	23.626550100	127.0.0.1	127.0.0.1	TLSv1.3	362	Application Data
120	23.626631700	127.0.0.1	127.0.0.1	TCP	44	55860 → 3443 [ACK] Seq=3159 Ack=1663 Win=63744 Len=0
131	28.641066600	127.0.0.1	127.0.0.1	TLSv1.3	68	Application Data
132	28.641115700	127.0.0.1	127.0.0.1	TCP	44	3443 → 55860 [ACK] Seq=1663 Ack=3183 Win=62208 Len=0
133	28.641178600	127.0.0.1	127.0.0.1	TCP	44	55860 → 3443 [FIN, ACK] Seq=3183 Ack=1663 Win=63744 Len=0
134	28.641210100	127.0.0.1	127.0.0.1	TCP	44	3443 → 55860 [ACK] Seq=1663 Ack=3184 Win=62208 Len=0
135	28.641458000	127.0.0.1	127.0.0.1	TLSv1.3	68	Application Data (Encrypted POST /api/signup)
136	28.641522300	127.0.0.1	127.0.0.1	TCP	44	55860 → 3443 [RST, ACK] Seq=3184 Ack=1687 Win=0 Len=0
345	44.476077200	127.0.0.1	127.0.0.1	TCP	56	52756 → 3443 [SYN] Seq=0 Win=65535 Len=0 MSS=65495
346	44.476136400	127.0.0.1	127.0.0.1	TCP	56	3443 → 52756 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
347	44.476198900	127.0.0.1	127.0.0.1	TCP	44	52756 → 3443 [ACK] Seq=1 Ack=1 Win=65280 Len=0

> Frame 135: Packet, 68 bytes on wire (544 bits), 68 bytes captured
> Null/Loopback
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 3443, Dst Port: 55860, Seq: 1663
> Transport Layer Security (TLSv1.3 Application Data: Opaque Ciphertext)

```
0000 02 00 00 00 45 00 00 40 65 54 40 00 80 06 97 61   ....E..@eT@....a
0010 7f 00 00 01 7f 00 00 01 0d 73 da 34 51 cb 2e 17   .........s.4Q...
0020 ee 24 77 5c 50 18 00 f3 3d bd 00 00 17 03 03 00   .$wP...=.......
0030 13 b3 4b 71 54 52 1c 21 1d e2 0f 91 34 f2 d6 5c   ..KqTR.!....4..
0040 e2 47 a1 51                                       .G.Q
```

Figure 3: signup capture

Login page

The login page accepts a username or the registered email together with the password and submits them to /api/login.

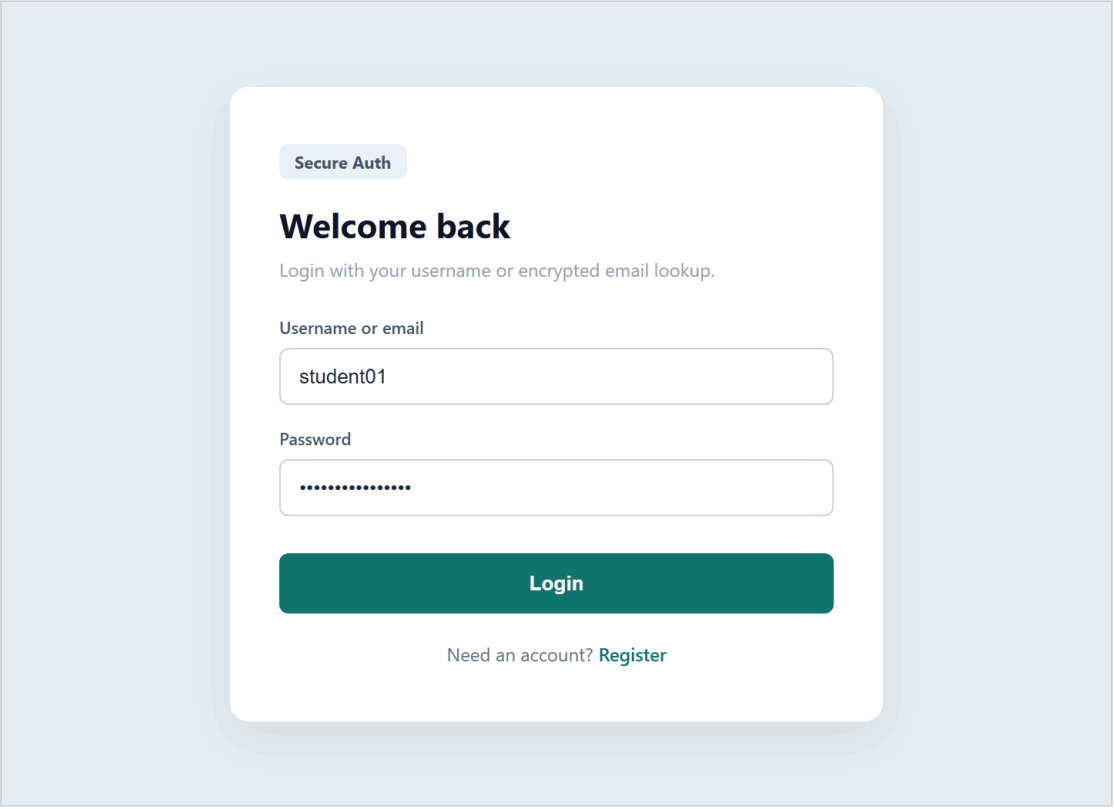


Figure 4:Login Page

Filter: `tcp.port == 3443`

No.	Time	Source	Destination	Protocol	Length	Info
847	151.378775000	127.0.0.1	127.0.0.1	TCP	44	58682 → 3443 [ACK] Seq=3156 Ack=2152 Win=62208 Len=0
850	151.455797000	127.0.0.1	127.0.0.1	TLSv1.3	758	Application Data (POST /api/login)
880	151.456074000	127.0.0.1	127.0.0.1	TLSv1.3	68	Application Data
881	151.577181000	127.0.0.1	127.0.0.1	TCP	44	58682 → 3443 [ACK] Seq=3157 Ack=2910 Win=62208
882	151.577284000	127.0.0.1	127.0.0.1	TCP	44	58682 → 3443 [RST, ACK] Seq=3157 Ack=2910 Win=0
884	151.577312000	127.0.0.1	127.0.0.1	TLSv1.3	372	Application Data (Encrypted Credentials Response)
890	151.588102000	127.0.0.1	127.0.0.1	TCP	44	3443 → 58682 [ACK] Seq=2910 Ack=3158 Win=63744
895	151.599124000	127.0.0.1	127.0.0.1	TLSv1.3	880	Application Data
900	151.601244000	127.0.0.1	127.0.0.1	TLSv1.3	238	Application Data

> Frame 884: Packet, 68 bytes on wire (544 bits), 68 bytes captured
> Null/Loopback
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 3443, Dst Port: 58682, Seq: 2910
> Transport Layer Security (TLSv1.3 Application Data: Zero Plaintext)

```
0000 02 00 00 00 45 00 00 40 67 12 40 00 80 06 95 a7   ....E..@g.@....
0010 7f 00 00 01 7f 00 00 01 0d 73 da 34 51 cb 2e 17   .........s.4Q...
0020 ee 24 77 5c 50 18 00 f3 3d bd 00 00 17 03 03 00   .$w\P...=.......
0030 13 b3 4b 71 54 52 1c 21 1d e2 0f 91 34 f2 d6 5c   ..KqTR.!....4..\
0040 94 12 ee 4f                                       ...O
```

Figure 5: Login capture

Functional Test Results

The authentication flow was exercised with HTTPS requests to https://localhost:3443. Every case produced the expected result.

#	Test	Expected	Result
1	Signup with invalid email "not-an-email"	Rejected	400: A valid email address is required
2	Signup with 5-character password	Rejected	400: Password must be at least 12 characters
3	Signup with valid data	Account created	201: Account created
4	Signup again with same username and email	Rejected	409: Username or email already registered
5	Login with wrong password	Rejected	401: Invalid credentials
6	Login with username and correct password	Success, cookie set	200: Login successful; cookie HttpOnly, Secure, SameSite=Strict
7	Login with email (different letter case)	Success	200: Login successful (blind index match)
8	Read profile with session cookie	Email decrypted for owner	200: student01, student@example.com
9	Read profile without cookie	Denied	401: Not authenticated
10	Plain HTTP request to port 3443	No response	Connection fails (HTTP code 000)
11	TLS negotiation	TLS 1.3	TLSv1.3, TLS_AES_256_GCM_SHA384, X25519, certificate verified